Premise Design Prior to Linguistic Projection: An Independent Definition of the Premise Extraction Layer Based on Semantic Fields, Phase Preservation, and Re-injectability

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Abstract

This paper critiques the design assumption that natural language is the final carrier of thought, premises, and meaning, and instead defines an independent theoretical object that exists prior to linguistic expression: the *premise extraction layer*. The argument begins from the recognition that premises do not originally exist as propositional strings, but as semantic fields containing high-dimensional constraint structures, responsibility traces, contextual bindings, exception conditions, and unspoken suspended components. Natural language, by contrast, projects these structures into one-dimensional linear sequences and therefore unavoidably introduces compression distortion, order distortion, granularity collapse, and contextual stripping. Consequently, if one wishes to design an OS-like structure capable of premise updating and high-density thought operation, one must define a layer that preserves, extracts, and reprojects pre-linguistic semantic structure before attempting to improve language itself.

This paper defines that layer as the *Premise Extraction Layer* and presents six basic components: (i) semantic field, (ii) constraint set, (iii) phase-preserving representation, (iv) linear linguistic projection, (v) re-injectability, and (vi) semantic density preservation ratio. It further argues that a premise is not a summary of text but an extraction of operative constraints; that natural language is not the premise itself but a purpose-specific view of a phase-preserving representation; and that silence, suspension, and indeterminacy are not deficiencies but constitutive elements of semantic structure. The paper concludes that to design *beyond language* does

not mean escaping into the non-linguistic; rather, it means demoting language to one final output form and fixing the semantic field itself as the higher-order object of design.

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1. Introduction

1.1 Problem Setting

In advanced thought, creativity, institutional design, dialogue, and AI operation, what matters is often not merely *what is said*, but *under which premises it is said*. Yet such premises have typically been treated under vague labels such as beliefs, values, positions, opinions, or worldviews, rather than as implementable structures. As a result, changes in premises are approximated as revisions of wording or modifications of explanation, while the operative structure of the premise itself remains untreated.

The problem addressed in this paper is straightforward: do premises truly exist as sentences? If not, in what form do premises exist, and how are they projected into language and then returned to operation? This paper approaches that question not as a theory of language alone, but as a theory of premise design.

1.2 Main Claim

The central claim of this paper may be summarized in a single sentence:

> A premise does not exist as a natural-language sentence; it exists as a constraint structure embedded in a semantic field, and natural language is only the result of projecting that structure into a purpose-dependent linear form.

From this standpoint, the central problem of language design shifts away from producing better sentences or inventing new vocabularies. Instead, it becomes the problem of how to preserve pre-linguistic semantic structure, project it with minimal loss, and return it back into an operative system.

1.3 Purpose of This Paper

The aim of this paper is to independently define the following:

1. To redefine premises not as propositional strings but as constraint sets
2. To introduce the semantic field as the layer in which premises actually exist
3. To position natural language as a linear projection of a semantic field
4. To define a phase-preserving representation required prior to linguistic projection
5. To evaluate linguistic expression through re-injectability and semantic density preservation

2. The Basic Problem: Why Language Alone Is Not Enough

2.1 A Premise Is Not a Sentence

When a person makes a judgment, multiple constraints operate simultaneously in the background. For example, the judgment "authenticity should be prioritized over efficiency" is not merely a sentence. It contains at least the following:

- what counts as efficiency,
- what counts as authenticity,
- how priority is resolved when the two conflict,
- under what conditions exceptions are allowed,
- in what context the judgment is valid,
- and what history produced the judgment.

Therefore, although a premise may be *expressed* in a sentence, it does not *exist* as a sentence. It exists as a constraint structure with multiple dependency relations.

2.2 Natural Language Is a Linear Mapping

Natural language is output as a one-dimensional sequence with temporal order. At that point, structures that originally coexisted simultaneously are rearranged into symbolic succession. This generates at least four distortions:

(a) **Compression distortion**: multi-layered constraints are compressed into one sentence

(b) **Order distortion**: elements that originally coexist acquire artificial before/after relations

(c) **Granularity distortion**: root principles and local rules appear on the same level

(d) **Context stripping**: scopes of application and exception conditions fall away

Thus, natural language is not only a medium for conveying meaning; it is also a medium that can deform meaning.

2.3 Summarization and Extraction Are Different

The most common mistake in handling premises is to conflate summarization with extraction. Summarization reduces the number of sentences; extraction identifies structure. The goal of summarization is brevity, whereas the goal of extraction is to reveal the effective constraints that govern operation. Accordingly, premise design requires not a text summarizer, but a constraint extractor.

3. Notation and Minimal Assumptions

Let i denote an agent and t denote time. The following minimal notation is used:

- $\Psi_i(t)$: semantic field of agent i
- $C_i(t)$: premise constraint set of agent i
- $\Phi_i(t)$: phase-preserving representation
- $L_i^{(\omega)}(t)$: linguistic projection for purpose ω
- $\mathcal{P}$: linear projection operator into language
- $\mathcal{Q}$: re-injection (re-parsing) operator from language
- $\rho(\cdot)$: semantic density
- $R_i(t)$: reprojection history of premise updates

Here, ω denotes usage purpose, such as internal operation, external explanation, education, documentation, signature, or institutional connection.

4. Definition of the Premise Extraction Layer

4.1 Definition of the Semantic Field

Definition 1 (Semantic Field). The semantic field $\Psi_i(t)$ of agent i is a high-dimensional semantic configuration in which judgments, utterances, suspensions, refusals, expectations, responsibility traces, and contextual constraints are superposed. It does not exist as a set of sentences but as a distribution of possibilities and constraints.

The semantic field contains not only what has been expressed, but also what has not been expressed, what has not yet been formalized, and what remains suspended as unresolved tension. Hence, the semantic field is not a completed knowledge base, but an updatable structural field.

4.2 Definition of the Premise Constraint Set

Definition 2 (Premise Constraint Set). The premise constraint set $C_i(t)$ is the set of constraints extracted from the semantic field $\Psi_i(t)$ that actually govern judgment and connection, and is written as

$$C_i(t) = \{c_{i1}(t), c_{i2}(t), \ldots, c_{in}(t)\}.$$

Each constraint c_{ik} possesses at least the following attributes:

(i) scope,

(ii) priority,

(iii) exception conditions,

(iv) update conditions,

(v) evidential trace,

(vi) suspended components.

The crucial point here is that a premise is not a declaration such as "this is what I think," but an operative constraint governing what is permitted, refused, or kept in suspension under given conditions.

4.3 Premise Extraction Operator

Definition 3 (Premise Extraction). Premise extraction is the operation that derives the premise constraint set $C_i(t)$ from the semantic field $\Psi_i(t)$ and update history $R_i(t)$:

$$\mathcal{E}_i : (\Psi_i(t), R_i(t)) \mapsto C_i(t).$$

This extraction should take into account not merely the surface frequency of uttered content, but also repeated selections, refusal tendencies, silences, and revision histories, because a person's premises appear more reliably in traces than in explanations.

4.4 Definition of the Phase-Preserving Representation

Definition 4 (Phase-Preserving Representation). The phase-preserving representation $\Phi_i(t)$ is an intermediate representation that preserves the dependency relations, hierarchy, conflicts, exceptions, and scopes of the constraint set $C_i(t)$ without collapsing them:

$$\Phi_i(t) = \mathcal{M}(C_i(t)).$$

Here, $\mathcal{M}$ may take the form of a graph, tensor, directed constraint network, hierarchical bundle, or any other structure-preserving representation. What matters is not the specific formalism, but the requirement that structure be preserved before being projected into natural language.

4.5 Definition of Linear Linguistic Projection

Definition 5 (Linear Linguistic Projection). For a given purpose ω, the natural-language output $L_i^{(\omega)}(t)$ is defined as

$$L_i^{(\omega)}(t) = \mathcal{P}_\omega(\Phi_i(t)),$$

where $\mathcal{P}_\omega$ is an operator that projects the phase-preserving representation into a purpose-dependent one-dimensional sequence.

Under this definition, natural language is not the premise itself, but a purpose-specific view of the premise.

4.6 Definition of Re-injectability

Definition 6 (Re-injectability). A linguistic projection $L_i^{(\omega)}(t)$ is re-injectable if, through the re-parsing operator $\mathcal{Q}$,

$$\mathcal{Q}(L_i^{(\omega)}(t)) \approx \Phi_i(t)$$

holds to a sufficient degree.

Here, "$\approx$" does not imply complete identity. However, at minimum, root constraints, priority relations, exception conditions, and suspended components must not be fatally lost.

4.7 Semantic Density Preservation Ratio

Definition 7 (Semantic Density Preservation Ratio). As an index of how much of the original semantic structure is preserved through linguistic projection, define the semantic density preservation ratio $\eta_i^{(\omega)}$ by

$$\eta_i^{(\omega)} = \frac{\rho(\mathcal{Q}(L_i^{(\omega)}(t)))}{\rho(\Phi_i(t))}.$$

Here, ρ is an abstract quantity representing the density of semantic constraints or dependency relations retained in a representation. Ideally, $\eta_i^{(\omega)} \to 1$, though in practice each usage purpose will have its own acceptable loss threshold.

5. Axiomatic Conditions of the Premise Extraction Layer

5.1 Axiom 1: Premises Are Constraints, Not Propositions

Axiom 1. A premise exists not as a single proposition subject to truth evaluation, but as a set of constraints governing connection, selection, refusal, and suspension.

Under this axiom, differences in premises are treated not as differences in opinion, but as differences in the rules that generate operative space.

5.2 Axiom 2: Natural Language Is a Projection, Not the Substance

Axiom 2. A natural-language sentence is not the substance of a premise, but only a linear projection of a phase-preserving representation. Therefore, one cannot assume that a sentence alone fully recovers the premise itself.

5.3 Axiom 3: Silence and Suspension Are Not Deficiencies

Axiom 3. Unexpressed content, silence, suspension, and indeterminacy are not missing data but constitutive elements of the semantic field.

This axiom is crucial. In premise design, what has not yet been said, what has not yet been defined, and what is deliberately left unsettled are not defects; they are operational margins that preserve reversibility.

5.4 Axiom 4: Extraction Takes Priority over Summarization

Axiom 4. In premise operation, extraction of constraint structure takes precedence over textual simplification.

This means that a highly readable explanatory sentence is not necessarily a high-fidelity premise representation.

6. Propositions and Corollaries

6.1 Proposition 1: A Design Without a Phase-Preserving Layer Damages Premises

Proposition 1. A design that projects directly from the semantic field into natural language, without passing through a phase-preserving representation $\Phi_i(t)$, is prone to damaging dependency structures within the premise.

Reason. The semantic field $\Psi_i(t)$ is high-dimensional, whereas natural language is

one-dimensional. A direct projection therefore induces structural degeneration. By introducing a phase-preserving layer, one can at least localize and track structural loss.

6.2 Proposition 2: A Non-Reinjectable Language Cannot Serve as an Operative Language

Proposition 2. A linguistic expression lacking re-injectability may function as an explanation, but it is insufficient as an internal operational language of a premise-updating OS.

Reason. Internal operation requires that linguistic expressions be recoverable into judgment conditions, update conditions, and log differences. Expressions that cannot be parsed back may function as essays or speeches, but they cannot be returned into an updating machine.

6.3 Proposition 3: Different Purposes Require Different Linguistic Projections

Proposition 3. From the same phase-preserving representation $\Phi_i(t)$, one requires multiple linguistic projections for different purposes:

$$L_i^{(\omega_1)}(t), L_i^{(\omega_2)}(t), \ldots$$

Corollary 1. Accordingly, the relation between an "original text" and its "translation" should not be understood as that between an authentic original and a degraded copy, but as that between different reprojections of a common underlying structure.

6.4 Proposition 4: Premise Updating Is Not Sentence Revision but Constraint Reorganization

Proposition 4. Premise updating is not the rewriting of sentences, but the reorganization of the constraint set and phase representation:

$$C_i(t) \mapsto C_i(t+1), \qquad \Phi_i(t) \mapsto \Phi_i(t+1).$$

Corollary 2. Therefore, a true update log should not be a sentence-difference log, but a constraint-difference log and a phase-difference log.

7. Design Conditions

From the theory developed in this paper, the following minimum conditions for premise design prior to linguistic projection are derived:

1. Premises must be stored as constraint sets rather than sentences
2. Constraint sets must be mapped into phase-preserving representations
3. Natural-language expression should occur only as purpose-specific linear projection
4. Linguistic expressions must be re-injectable
5. Silence, suspension, and indeterminacy must be preserved as states
6. Updates should be tracked as constraint differences and phase differences rather than sentence differences
7. Evaluation criteria must include semantic density preservation, not readability alone

8. Implications for AI, Education, and Institutional Design

8.1 Implications for AI

Many current AI systems depend heavily on correspondences between input sentences and output sentences, or on predictive accuracy over token sequences. From the standpoint of this paper, such systems lack the layer prior to phase-preserving representation. If AI is to support premise updating or high-density dialogue, what is required is not merely a sentence generator, but a structure capable of:

- estimating semantic fields,
- extracting constraint sets,
- preserving phase representations,
- and generating re-injectable linguistic projections.

8.2 Implications for Education

Many failures in education arise from compressing the learner's premise structure too quickly into textual answers. Questions, suspensions, silences, and indeterminacies are not deficiencies; they are the margin within which premises are formed. Therefore, what education needs is not answer formulation alone, but a design that protects the premise extraction layer prior to linguistic fixation.

8.3 Implications for Institutional Design

Institutions often fix complex premise structures into legal clauses, rules, and commands. But once externalized, such fixation carries high irreversibility. Therefore, unless a sufficiently rich phase-preserving representation is maintained before externalization, institutional language risks becoming a crude compression of the underlying premise.

9. Discussion: What Does It Mean to Design Beyond Language?

9.1 A Misunderstanding of "Beyond Language"

The phrase "designing beyond language" is often misunderstood as an appeal to non-linguistic intuitionism or mysticism. That is not the position of this paper. Here, to go beyond language does not mean abandoning language, but refusing to treat language as the sole object of design.

9.2 Shift in the Object of Design

The conventional design sequence is often:

$$thought \rightarrow sentence \rightarrow transmission \rightarrow interpretation.$$

By contrast, this paper proposes:

$$semantic\,field \rightarrow constraint\,extraction \rightarrow phase\,preservation \rightarrow linguistic\,projection \rightarrow re\text{-}injectio$$

In this sequence, natural language becomes not the center but a relay layer. The center of design shifts to semantic fields and premise structures.

9.3 Repositioning Language

Under this framework, natural language does not become unnecessary. Rather, it is repositioned as a necessary projection layer for socially connecting higher-order structure with loss:

- sharing internal structure,
- external explanation,
- minimal signature,
- institutional connection,

- educational reprojection,
- dialogical resonance.

Thus natural language is not "something ultimately dispensable," but "a necessary projection layer that connects higher-order structure to society under controlled loss."

10. Future Tasks

This is a definitional paper and leaves the following open:

1. concrete measurement methods for semantic density ρ,
2. formal specifications for phase-preserving representation Φ,
3. quantitative criteria for re-injectability,
4. implementation methods for constraint-difference logging,
5. synchronization conditions among multiple agents' semantic fields,
6. and pre-externalization compression-loss diagnostics for institutions.

These form independent research tasks for implementation, verification, and operation.

11. Conclusion

This paper has defined premises not as sentences but as constraint structures embedded in semantic fields, and has positioned natural language as their purpose-dependent linear projection. It has further proposed the premise extraction layer as an independent theory: a layer that extracts constraint sets from semantic fields, passes them through phase-preserving representations, projects them into language, and returns them to operation.

The conclusion of this paper may be stated as follows:

> To design beyond language does not mean abandoning language. It means relocating the substance of premises from natural-language sentences to semantic fields, constraint structures, and phase preservation, and repositioning language as one final output form among others.

In this sense, the core of premise updating and high-density thought operation lies not first in *how to say something*, but in *what is preserved at which layer*. This paper has fixed that layer as the *premise extraction layer*.

Acknowledgment This paper was written as part of an ongoing line of thought concerning language, meaning, premises, and updatable structure.

12. Supplementary Proposition: Connection Between Implicit Internal Representation and the Premise Extraction Layer

Proposition 1 (Connection Between Implicit Internal Representation and Premise Extraction)**.** *The implicit internal representation R_{imp} that appears in high-density states such as dreams, intuitions, insights, and immediate post-awakening cognition is a nonlinear, parallel, and semantically dense structure that precedes natural-language sentences, and may therefore be positioned as a special phase of the semantic field Ψ within the premise extraction layer.*

Accordingly, when linguistic expression is performed directly through the sequential projection operator

$$L : R_{\mathrm{imp}} \to R_{\mathrm{exp}},$$

the resulting projection loss is large not because the internal representation is insufficiently formed, but because the intermediate extraction layer—namely, the passage through the constraint set C and the phase-preserving representation Φ—has been omitted.

Corollary 1. *The premise extraction layer does not deny the collapse associated with sequential projection from nonlinear semantic structures; rather, it presupposes such collapse and provides the structural conditions for making projection loss visible, localizable, and partially reducible.*

Explanation Under this proposition, the linguistic collapse of implicit internal representation should not be interpreted as evidence that meaning was absent or underdeveloped. Instead, it should be understood as the result of excessive compression caused by direct projection from a semantic field into natural language. The premise extraction layer responds to the sequential projection problem at the level of design by introducing intermediate stages of constraint extraction and phase preservation.